

Methodology: The CIF-AD-OODA Integration Model

This section establishes the analytical framework used to evaluate cognitive integrity threats across the ten critical domains examined in this paper. We integrate three complementary theoretical frameworks—the Cognitive Integrity Framework (CIF) [?], Axiomatic Design (AD) [?, ?], and the OODA Loop [?—into a unified model for analyzing Goal Hijacking attacks and their defenses.

Framework Integration

Each framework contributes a distinct analytical dimension:

The Cognitive Integrity Framework (CIF) provides the defense mechanism vocabulary. Developed formally in Paper 1 of this series [?] and validated computationally in Paper 2 [?], CIF defines five canonical defense mechanisms that protect agent cognitive states $\sigma_i = \langle \mathcal{B}_i, \mathcal{G}_i, \mathcal{J}_i, \mathcal{H}_i \rangle$ against adversarial manipulation. These mechanisms operate at the architectural, runtime, and coordination layers to maintain belief consistency, goal alignment, and provenance verifiability.

Axiomatic Design (AD) provides the structural model. Suh's